

	SYLLABUS FOR		
	EE 201 – CIRCUIT THEORY I		
	TED UNIVERSITY		
	DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING		
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Course Code	EE 201	Course Title	Circuit Theory I	
Type of Course	☒ Compulsory ☐ Elective	Semester	☒ Fall ☐ Spring ☐ Summer	
Credit Hours / ECTS credits	(3+0+2) 4 Credits, 8 ECTS	Pre-requisite	MATH 101 AND PHYS 101 (or PHYS 105)	
		Co-requisite	NONE	
Mode of Delivery	☒ Face-to-face ☐ Distance learning	Language of Instruction	☒ English ☐ Turkish	
Course Level	☒ Undergraduate ☐ Graduate (MS) ☐ Graduate (PhD)	Year of Study	☐ First Year ☒ Sophomore ☐ Junior ☐ Senior	☐ Masters ☐ Doctorate

Course Coordinator	Asst. Prof. Serkan Sahin E-mail: serkan.sahin@tedu.edu.tr Room: B147 Phone: +90 (312) 585 0542		
Office Hours	Mondays 14:00–16:00		
Teaching Assistant(s)	Zeynep Sıdıka Seven, E-mail: zeynep.seven@tedu.edu.tr , Room: A420		
Office Hours	To be determined		
Course Schedule	Section 01	Monday (11:00–13:00) @G208, Tuesday (11:00–12:00) @G208, Wednesday (11:00–13:00) @A317-L	

Catalog Description	Fundamentals of electric circuits, variables, and lumped circuit elements. Kirchhoff's laws. Resistive circuits. Methods of circuit analysis. Operational amplifiers (op amps). Energy storage elements. Analysis of first and second-order circuits. Experiments on resistive circuits, first and second-order circuits, and op amp circuits.
Course Objectives	This course equips students with the fundamentals of electric circuits, including energy storage, transient and steady-state responses, and operational amplifiers. The course covers the fundamental laws and techniques, such as nodal and mesh analysis and Thevenin and Norton equivalent circuits.
Required Reading	Nilsson, J.W., & Riedel, S. (2011). <i>Electric Circuits</i> . 9 th Ed., Prentice-Hall.
Suggested / Recommended Reading	• Alexander, C. K., & Sadiku, M. (2009). <i>Fundamentals of Electric Circuits</i>. 4th Ed., McGraw Hill. • Hayt, K., Kemmerly, J., & Durbin, S. (2018). <i>Engineering Circuit Analysis</i>. 9th Ed., McGraw-Hill Education. • Johnson, D. E., Johnson, J. R., Hilburn, J. L., & Scott, P. D. (1997). <i>Electric Circuit Analysis</i>. 3rd Ed., John Wiley & Sons.
Software Usage	MATLAB and LTspice
Course Learning Outcomes (LOs)	Upon successful completion of the course, students will be able to: 1. Recognize the basic circuit concepts, including voltage, current, energy, and power, 2. Express the node and mesh analysis methods used in linear time-invariant circuits analysis,

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	3. Apply Thevenin and Norton theorems for investigating circuits, including operational amplifiers, 4. Analyze alternating current (AC) circuits using sinusoidal steady-state analysis techniques, 5. Evaluate the behavior of capacitors and inductors in circuits, assessing both transient and steady-state responses of first-order circuits, 6. Experiment on resistive circuits, first and second-order circuits, and op amp circuits in collaboration with peers.
Course Evaluation	Course feedback survey will be conducted in the last two weeks of the semester.
Course Web Page	The class has already been enrolled in Moodle (https://lms.tedu.edu.tr/). All announcements and course-related materials will be posted on the Moodle course page.
Some Useful Website References	N/A

Learning Activities & Teaching Methods	☒ Brainstorming ☐ Case Study/Scenario Analysis ☒ Collaborating ☐ Concept Mapping ☒ Demonstrating ☒ Discussions / Debates ☐ Drama / Role Playing ☒ Experiments ☐ Field Trips ☐ Guest Speakers	☒ Hands-on Activities ☒ Inquiry ☐ Microteaching ☐ Oral Presentations / Reports ☐ Peer Teaching ☐ Predict-Observe-Explain ☒ Problem Solving ☒ Questioning ☒ Reading	☐ Scaffolding / Coaching ☐ Seminars ☐ Service Learning ☐ Simulations & Games ☒ Telling / Explaining ☐ Think-Pair-Share ☐ Video Presentations ☒ Web Searching ☐ Other(s):
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Assessment Methods & Criteria	☐ Case Studies / Homework	(...%)	☐ Presentation (Oral, Poster)	(...%)
	☒ Lab Assignment	(20%)	☐ Project	(...%)
	☐ Observation	(...%)	☒ Quiz	(10%)
	☐ Oral Questioning	(...%)	☐ Self-evaluation	(...%)
	☐ Peer Evaluation	(...%)	☒ Test/Exam	(70%)
	☐ Performance Project (Written, Oral)	(...%)	☐ Other(s):	(...%)
	☐ Portfolio	(...%)		

Student Workload	☐ Case Study Analysis	(... hrs)	☐ Online Discussion	(... hrs)
	☒ Course Readings	(75 hrs)	☐ Oral Presentation	(... hrs)
	☐ Debate	(... hrs)	☐ Poster Presentation	(... hrs)
	☐ Demonstration	(... hrs)	☒ Report on a Topic	(39 hrs)
	☒ Exams/Quizzes	(56 hrs)	☐ Research Review	(... hrs)
	☐ Field Trips/Visits	(... hrs)	☐ Resource Review	(... hrs)
	☐ Hands-on Work	(... hrs)	☐ Team Meetings	(... hrs)
	☒ Lab Applications	(28 hrs)	☐ Web Designs	(... hrs)
	☒ Lectures	(42 hrs)	☐ Work Placement	(... hrs)

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	☐ Mock Designs	(... hrs)	☐ Workshop	(... hrs)
	☐ Observation	(... hrs)	☐ Other(s):	(... hrs)
	Total Workload			240 hrs

TENTATIVE OUTLINE				
<i>Any changes and updates will be announced on the course web page.</i>				
Week	Topics	LOs	Textbook Reading	Assignments
1	Introduction to Electrical Circuits	1	Ch. 1	
2	Resistive Circuits; Sources; measurement equipments	1	Ch. 2, 3	
3	Linearity; Nodal Analysis	2	Ch. 3, 4	Quiz#1
4	Nodal Analysis; Mesh Analysis	2	Ch. 4	
5	Mesh Analysis Lab#1: Intro. to Elec. Circuit Laboratory	2	Ch. 4	Lab#1
6	Thevenin's and Norton's theorems; Power Transfer; Superposition Lab#2: Intro. to Mixed Signal Oscilloscope and Waveform Generator	3	Ch. 4	Quiz#2 Lab#2
7	Op-Amps	4	Ch. 5	Midterm Exam
8	Analysis of resistive Op-Amp circuits Lab#3: The DC and AC Analysis on Resistive Circuits	4	Ch.5	Lab#3
9	Energy-Storage Elements	5	Ch. 6	Quiz#3
10	First-Order Circuits Lab#4: Operational Amplifiers	5	Ch. 7	Lab#4
11	First-Order Circuits	5	Ch. 7	
12	Second-Order Circuits Lab#5: Capacitors, Inductors, and First Order Circuits	6	Ch. 8	Lab#5
13	Second-Order Circuits	6	Ch. 8	Quiz#4
14	Sinusoidal Steady-State Analysis Lab#6: Second Order Circuits	7	Ch. 9	Lab#6
FINAL EXAMS WEEK (date and time to be announced later).				

COURSE ASSIGNMENTS & GRADING
A. Midterm Exams [30%]
There will be one midterm exam. The exam will be in-class. The date and how it is taken will be announced later.



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B. Final Exam [40%]

There will be a cumulative final exam covering all topics. You are allowed to use cheat sheets in the exam. The date and time of the final exam will be announced at the end of the semester.

C. Quizzes [10%]

There will be 4 quizzes, 2.5% for each quiz. The method of conducting the quizzes will be explained later.

D. Lab Assignments [20%]

There will be 6 experiments, each contributing 3.33% to the overall grade. For each experiment, you will be required to prepare a prelab report, a final report, and attend a post quiz.

Lab Grading: Lab Performance [10%], Preliminary reports [30%], Final lab reports [40%], Lab Quiz [20%]

COURSE ASSESSMENTS & LEARNING OUTCOMES MATRIX

Assessment Methods	Course Learning Outcomes
Quizzes	LO# 1–8
Lab Assignments	LO# 1–8
1 st Midterm Exam	LO# 1–3
2 nd Midterm Exam	LO# 4–7
Final Exam	LO# 1–7

COURSE POLICIES

I. Attendance

- Regular class attendance is expected for all students at the University. You are not required but advised to attend all classes.
- Please send your professor a brief e-mail to explain your absence in advance.
- Your absence will not reduce your attendance rate *if and only if* you have a legitimate reason for missing a class (such as illness, death in the family, a traffic accident, *etc.*).
- In case of a disease or emergency, you must supply formal documentation that supports your claim.
- Classes start at the hour. Please be respectful of your classmates by being on time.
- All electronic equipment should be turned off and kept out of sight before the lecture starts.

About the *laboratory sessions*:

- The students should attend all the experiments. Those who miss two experiments without any allowable excuse (medical report or coinciding exams) will fail the laboratory.
- You have only one right to be absent from experiments.
- Make-ups may only be given to those who have allowable excuses. If you have a medical report or a conflicting exam, you should consult your assistant with a written formal request and your request will be assessed whether you will be given a make-up.
- Students who missed lab quizzes (including make-ups) will not be allowed to take lab quizzes.

II. Make-up Policy (for Exams and Assignments)

There will be a single make-up exam for the Midterm Exam which will be available if and only if you have a legitimate reason for missing the exam (such as illness, death in the family, a traffic accident, *etc.*). In case of an illness or emergency, you must supply formal documentation that supports your claim. The topics for the make-up exam are from everything that is covered in class at the time of the exam. There will be *no* make-up for the quizzes.

III. Preliminary Lab Report



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Preliminary work occurs from reading assignments, numerical calculations, and computer simulations. Each student should prepare his/her preliminary work and submit it at the beginning of the experiment. **The works should be neatly written on A4 white paper.** The graphs should be plotted on a computer and past to the A4 paper. The outcomes of computer simulations should be added to the preliminary work.

☐ **Your preliminary reports should be limited to a maximum of 5 pages without a cover page. Only those students who have submitted the preliminary work at the beginning of the laboratory will be allowed to experiment!**

IV. Lab Quizzes

A quiz will be given at the beginning of each experiment (except the introductory experiments at the beginning of the semester). The quiz questions will cover the subjects included in the preliminary works (including the reading assignments).

V. Equipment

Each team has to bring a **scientific calculator and a flash disk**. The calculator may be used during the quiz and the experiment. The students are responsible for the equipment on their experiment table. The equipment should be checked at the beginning of each laboratory session and in case of any missing equipment, the students should inform their assistants. Exchanging of equipment is not allowed among the tables without the permission of the assistant. **At the end of each experiment, one of the datasheets (if available) (last page of the lab manual) must be submitted to the course assistant.** The other datasheet must be kept by the students and used for the final report.

VI. Final Lab Report

A final report about the experiment should be prepared after the experiment and **submitted before the next laboratory session.** (Deadline for the last experiment will be announced by the assistant.) The reports should be neatly written on A4 white paper and prepared via computer. The graphs should be added to this paper. Each team of two students will submit one final report including:

- Name of the students
- Name of the assistant
- Date
- Name of the experiment
- The objective of the experiment
- Graphical results, measurements, and answers to the questions
- Comparison of the results with the expected ones and comments
- Conclusions

Please see the following parts of this document to learn how to prepare a reasonable report. **Your final lab reports should be limited to a maximum of 5 pages without a cover page.**

VII. Late Submission Policy

Late submissions will not be graded. Missed quizzes will result in a grade of zero (0).

VIII. Cheating & Plagiarism

Collaboration is strongly encouraged; however, the work you hand in must be solely your own. Cheating and plagiarism are severe offenses and will be penalized accordingly by the university disciplinary committee. Cheating has a comprehensive description which can be summarized as "acting dishonestly." Some of the things that can be considered cheating are the following:

- Copying answers on exams, home works and lab works,
- Using prohibited material on exams,
- Lying to gain any advantage in class,
- Providing false, modified, or forged data in a report,
- Plagiarizing (see below),
- Modifying graded material to be re-graded,
- Causing harm to colleagues by distributing false information about an exam, homework, or lab.

All of the following are considered plagiarism (www.plagiarism.org):

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- Turning in someone else's work as your own,
- Copying words or ideas from someone else without giving credit,
- Failing to put a quotation in quotation marks,
- Giving incorrect information about the source of a quotation,
- Changing words but copying the sentence structure of a source without giving credit,
- Copying so many words or ideas from a source that it makes up the majority of your work, whether you give credit or not.

IX. Disability Support

If you have a disabling condition that may interfere with your ability to complete this course successfully, please contact Dr. Serkan Sahin (e-mail: serkan.sahin@tedu.edu.tr). For more information, please see the [Handbook for Registered Students](#) (Turkish Content).

X. Student Development and Psychological Counseling Center

Student Development and Psychological Counseling Centre is mandated with providing crisis intervention and supportive listening services to the campus community. The Center conducts individual counseling, group guidance studies, workshops, seminars, and orientation studies for all students in need. You may apply to the Center in order to deal with all your current problems. For further information and/or questions: ogrencidanismamerkezi@tedu.edu.tr, <http://csc.tedu.edu.tr/>

XI. TEDU Without Barriers Unit

Please inform the TEDU Without Barriers Unit and the instructor of the course about the specific issues in case you have a physical or mental disability and are having trouble with anything related to this course—such as accessing the material, participating in the class, taking notes, preparing for, attending or managing to complete the exams. Your situation will be reviewed by commission, in accordance with the principle of confidentiality, and if deemed appropriate, facilitating measures will be taken so that you can take the course more efficiently. For further information and/or questions: engelsiz@tedu.edu.tr, <https://www.tedu.edu.tr/engelsiz-tedu>

Prepared By	Asst. Prof. Serkan Sahin	Date	03/10/2023
Revised By	Asst. Prof. Serkan Sahin	Rev. Date	03/10/2023

HOW TO PREPARE A FINAL LAB REPORT

Engineers, scientists, and managers write reports to communicate the results of research, fieldwork, and other activities. Often, a report is the only concrete evidence of your research, and the quality of the research may be judged directly by the quality of the writing and how well you convey the importance of your findings. Content, organization, and clarity are the keywords to remember when preparing a report.

Reporting the results obtained from an experiment is as important as experimenting. A sloppy report significantly deducts from the value of the results. Therefore, a properly prepared report is very important. Fortunately, a lab report has a fairly consistent format that will help you to organize your information. This document covers some important points on how to prepare a proper report.

The lab report must include:

- The title of the experiment
- The names and the ID numbers of the students (Upper left corner)
- The name of your teaching assistant (Upper left corner)
- The date that the lab was performed (Upper right corner)
- The lab section (Upper right corner)

There are three main sections of the laboratory report.

- Objective
- Results
- Conclusion

1) **Objective:**

You may describe the objectives of experiments in one or two sentences. The objective section in the report should not be copied directly from this section of the manual.

2) **Results:**

This section contains the results obtained in the experiment: **data tables and graphs.**

3) **Conclusion:**

You should compare your results against expectations from theory/simulation results, and quantify the differences. If the errors are significant, then **you should state the possible reasons for errors.** You should conclude the points of your results. The expression of results (graphs/data) mustn't be considered as a **“conclusion”**. Please, try to comment on results, not to narrate the experimental process.

Your reports should be limited to a maximum of 5 pages without a cover page. You should briefly summarize your results and conclusion.

Data Tables

Tables are effective and concise tools to summarize results and point to differences across parameters. It is a good habit to think of plan out the data you would like to get from an experiment by preparing tables before starting any measurement. The units should always be included along with the data.

Ali Velioglu 199999-9 Ahmet Mehmetoglu 200000-0 Assistant:	19-9-2010 Monday Morning
EXPERIMENT 1 Introduction to Laboratory Instruments Object Results Conclusion	

Figure 1: Lab report format.

Plotting the Graphs

Some of the results are presented by graphs more efficiently. Therefore, graphs are frequently used in reports. A proper plot should obey the following rules:

- The title of the plot should be written.
- Vertical and horizontal axes should be named.
- Units of the values should always be mentioned.
- Critical values should always be mentioned.

A few important points about the reports are as follows:

- You should write in complete, grammatically correct sentences.
- If most of your sentences are long (4 or more 'clauses' or parts) you will confuse the reader. Consider making two sentences (with 3 or fewer parts in each).
- Be concise. If you can use one word instead of a phrase with two or more words, then choose the one alternative.
- Be objective. Limit your use of personal pronouns (I, you, we), emotionally loaded words (wonderful, useless, lovely), and casual or ambiguous expressions.
- Use technical terms correctly. Learn what they mean, how to use them, and how to spell them.
- Do not use contractions (isn't, doesn't, it's), While these are common in speech, informal writing the full form (is not, does not, it is) is expected.
- Do not forget to indicate units of the values.

A good lab report does more than present the data; it demonstrates the writer's comprehension of the concepts behind the data. Merely recording the expected and observed results is not sufficient; you should also identify how and why differences occurred, explain how they affected your experiment, and shows your understanding of the principles the experiment was designed to examine. Bear in mind that a format, however helpful, cannot replace clear thinking and organized writing. You still need to organize your ideas carefully and express them coherently.